

Guided tongue reconstruction: feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model

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Abstract

Objective: Virtual Surgical Planning (VSP) for tongue reconstruction promises patient-specific optimizations, but its utility is limited by the difficulty of translating a digital plan to a deformable organ. This study evaluated the feasibility and anatomical accuracy of a patient-specific, additively manufactured surgical toolkit through benchtop experiments on soft-tissue phantoms. **Method:** Four distinct hemiglossectomy reconstruction cases were planned using VSP and executed on patient-specific silicone tongue phantoms. The guided workflow, performed by an independent surgeon, incorporated: (1) resection marking guide, (2) intraoperative volume inspection guide, (3) donor flap harvesting guide with decoupled planar and volumetric control, and (4) marker-guided flap inseting. High-resolution 3D surface scans were used to quantify anatomical fidelity by measuring the surface deviation between the pre- and post-procedure scans. **Results:** All four guided reconstructions were successfully completed. Mean surface deviation was 0.76–1.26 mm, and

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the 99th percentile (robust maximum) ranged 3.37–4.13 mm. This accuracy is comparable to that reported for established patient-specific osseous reconstructions. The operating surgeon found the toolkit intuitive, the workflow sequencing clear, and the volumetric inspection step practical. **Conclusion:** This benchtop study demonstrates that a patient-specific, guided surgical toolkit can feasibly and reliably translate tongue VSP into accurate reconstructions on soft-tissue phantoms. By introducing a volume feedback mechanism and decoupling the flap’s planar shape from its volume, the system provides adaptive guidance that is both geometry- and volume-aware. These findings support progression to cadaveric studies and early clinical validation to evaluate overbulking strategies, intraoperative adaptability, and correlation with functional outcomes.

Keywords: Virtual Surgical Planning, Patient-Specific 3D-Printed Surgical Guides, Free Flap Tongue Reconstruction, Surgical Simulation on Soft-Tissue Phantoms, Plastic and Reconstructive Surgery

1. Introduction

Virtual Surgical Planning (VSP) has become an established approach in head and neck reconstructive surgery [1]. It uses high-fidelity medical imaging and modeling to generate patient-specific, 3D-printed cutting guides, fixtures, and plates, reducing operative time, lowering costs, and improving functional outcomes [2, 3, 4, 5, 6, 7]. Building on this success, research has expanded to the complex domain of soft tissue reconstruction. For tongue reconstruction, VSP now allows surgeons to algorithmically design patient-specific soft tissue free flaps from a 3D model of the tongue and defect, yielding measurable clinical benefits such as reduced flap trimming, shorter operative times, and improved defect matching [8]. More advanced biomechanical models further refine this process by optimizing flap geometry to minimize tissue stretch and tension [9, 10].

While VSP for tongue reconstruction represents a significant leap forward, even an optimal digital plan remains a sophisticated visualization exercise unless it can be translated accurately to the surgical field [11]. This creates a critical “Execution Challenge”: the gap between the virtual design and surgical action. The innovation lies in implementing a guided surgical toolkit that serves as a bridge to transfer the digital blueprint to the operating room with high fidelity. This challenge is particularly pronounced for

deformable soft tissue, where the functional benefits of a biomechanically superior plan hinge on high-fidelity execution; geometric inaccuracy can negate the intended biomechanical advantages [10].

In mandibular and maxillofacial reconstruction, the clinical benefits often attributed to VSP arise from using a guided surgical toolkit to execute the plan with precision unattainable through freehand methods [12, 5]. The toolkit comprises instruments that translate the digital plan into precise, repeatable surgical actions. Core components include patient-specific, 3D-printed surgical guides for resection and osteotomy, patient-specific implants (PSIs) for fixation, and life-sized anatomical models for preoperative rehearsal and communication. The toolkit is a critical, often underappreciated, component that makes VSP clinically viable and transformative [5, 7].

Developing an effective surgical guiding toolkit for soft tissue reconstruction presents unique challenges. Unlike bone, a stable landmark-rich substrate for guide registration, soft tissues are mobile, deformable, and lack consistent reference points, making accurate transfer of a virtual plan far more complex [13]. Despite these hurdles, early work demonstrates the value of guided techniques across soft tissue applications. In autologous breast reconstruction, 3D-printed tools serve as anatomical guides for landmark identification and as physical molds for flap harvesting and shaping [14, 15]. Patient-specific templates derived from computed tomography angiography (CTA) can localize perforator vessels and significantly reduce intraoperative search time [16]. Guided accuracy has also been shown in complex facial procedures: sterilized contour guides for rhinoplasty translate the virtual plan with sub-millimeter accuracy [17]; multi-component systems for skin and cartilage improve symmetry in heminasal reconstruction [18]; and, in facial fat grafting, guides with integrated volumetric containers enable surgeons to precisely quantify and place the required tissue [19].

The tongue is complex, mobile, and deformable, and it lacks the stable bony landmarks used in osseous reconstruction, making high-fidelity translation particularly difficult. The literature reports several pioneering, albeit fragmented, attempts to bridge this virtual-physical gap, largely addressing two challenges: (1) guiding accurate resection of the complex three-dimensional tongue defect, and (2) guiding donor flap harvest in terms of planar geometry and volume.

Early VSP translation methods for tongue reconstruction relied on manually transferring landmark-based measurements from patient-specific images to guide both resection and flap harvest [20, 21]. More advanced approaches

use 3D printing to create physical resection guides; for example, patient-specific soft silicone guides that fit the tongue surface to delineate the planned resection line [22]. A key limitation of contact-based guides is achieving reliable registration on a mobile, deformable organ. Recent VSP workflows instead emphasize the second challenge: once the flap’s planar geometry is designed virtually, a physical template—printed on paper or 3D-printed—is sterilized and used during harvesting [8, 10]. Beyond geometry, VSP can also map perforator anatomy and tailor flap design to optimize vascularity [23].

Before VSP-based surgical guiding toolkits for tongue reconstruction can be clinically adopted, their feasibility and potential benefit must be evaluated in a controlled, non-clinical setting. Exploratory benchtop studies offer a platform for this initial validation. This approach was established in osseous reconstruction by Logan et al. [12], who conducted a benchtop study to evaluate surgical guides using 3D-printed fibula free flap and mandible models. While rigid anatomical models are sufficient for assessing osseous reconstruction, evaluating a toolkit for soft tissue demands higher material fidelity in the phantom; a realistic soft tissue analogue is needed to simulate the physical actions of cutting, handling, and suturing. Fortunately, several studies have focused on silicone phantom models for realistic skin [24] and human tongue [25, 26, 27].

To address the critical gap between virtual planning and execution in tongue reconstruction, we propose a patient-specific, additively manufactured surgical toolkit. This system is designed as a physical map that translates a biomechanically optimized virtual plan to the surgical field, reducing intraoperative guesswork and promoting procedural standardization. In this paper, we present the design details of this toolkit and demonstrate its feasibility and anatomical accuracy in an end-to-end workflow using soft-tissue phantoms. Accuracy is quantified from high-resolution 3D scans as pointwise surface deviation between the reconstructed neotongue and the reference pre-operative model.

2. Method

To address the challenges of developing a novel surgical system for complex soft tissue reconstruction, an iterative design framework was adopted. It was informed by its use in patient-specific design of facial prostheses [28] and grounded in established principles of guided surgical toolkits for mandibular and maxillary reconstruction. This strategy served as a de-risking step and

enabled rapid refinement of the surgical workflow through a continuous feedback loop with a senior surgeon co-author acting as the end-user evaluator. This granular, feedback-driven process shaped key aspects of the system’s development, from selecting fabrication methods to refining the surgical toolkit design, procedural workflow logic, clinical practicality, and intuitive integration into the procedure. As a result, engineering decisions were directly informed by clinical expertise and real-world procedural demands.

2.1. Tongue Reconstruction Virtual Surgical Planning

The feasibility of the proposed surgical toolkit was assessed in four clinically plausible tongue cancer cases. A previously published MRI-derived 3D tongue model from a healthy male participant in his early 40s (informed consent obtained from the Health Research Ethics Board of Alberta: HREBA.CC-17-0597) was reused without modification [10]. To span a representative spectrum of Urken class I–II tongue defects [29], four resections were computationally simulated as detailed previously [10]: tip-sparing hemiglossectomy, lateral hemiglossectomy, posteriorly extended hemiglossectomy (sparing the tongue base), and an asymmetric, midline-crossing partial glossectomy.

For each case, the patient-specific Virtual Surgical Plan (VSP) served as the toolkit’s foundation and was treated as a black box input for this study. The technical details of the VSP algorithm used to generate these plans are described elsewhere [9, 10]. For guided toolkit design, the VSP provides three essential 3D model outputs: the pre-operative tongue model (Figure 1B), the planned resection volume (Figure 2C), and the corresponding free flap design (Figure 3A and 3E). These digital models serve as the blueprints from which the patient-specific surgical toolkit is engineered.

2.2. The Patient-Specific Surgical Toolkit

The surgical toolkit was designed to address the primary challenge of translating a digital plan onto a mobile, deformable anatomy. To be effective, the guides had to be intuitive to use and to support stable, unambiguous registration using perimeter markers aligned to the patient’s surface topography. The toolkit provides physical constraints for surface marking and flap harvesting, while enabling volumetric control through a dedicated inspection guide.

The first phase of the guided procedure focuses on accurate defect resection. The surgeon begins by positioning the Resection Marking Guide (Figure 2A), a 2 mm-thick, shell-like instrument that conforms to the tongue

surface to delineate the planned incision boundary (Figure 2B). Placement is governed by distinct geometric indicators (e.g., circle, triangle, square) on the guide perimeter that correspond to soft-tissue anchor points selected during planning—such as the lingual apex and the junction of the median sulcus and circumvallate papillae. Aligning these perimeter markers with their anatomical counterparts enables plan-compliant positioning and physically constrains the tissue before suturing.

To aid visualization of cut depth and the complex three-dimensional shape, a solid Resection Model (Figure 2C) is also included. This physical replica of the removed tissue serves as a physical reference, helping the surgeon estimate the required depth of dissection. After resection, the tissue is placed into the Resection Volume Inspection Guide (Figure 2F). This tool functions as a semi-quantitative pass/fail gauge to assess volumetric fit [14]. By enclosing the tissue within the rigid two-part negative mold, the surgeon receives immediate feedback based on physical fit: resistance to mold closure signals a larger resection volume than planned, whereas closure with visible gaps indicates a deficit. This binary assessment provides a directional baseline, indicating whether the subsequent flap harvest should be scaled up or down relative to the plan.

The second part of the toolkit targets the donor site. It provides a two-part Flap Harvesting Guide that transfers the planned planar geometry and volumetric requirements to the donor site phantom. It consists of a planar template (Figure 3A) for outlining the skin perimeter and a volumetric guide (Figure 3B) to verify that the free flap volume matches the plan. This step is complemented by a 3D-printed Flap Model (Figure 3E), which serves as a physical reference, helping the surgeon internalize the complex target geometry and reducing the cognitive load of visualizing the volumetric requirements. Once harvested, the flap is transferred and inset into the neotongue. Correct orientation is achieved using the same system of geometric indicators (e.g., circle-to-circle) marked on the perimeters of both the resection and flap harvesting guides. These symbols map the flap’s periphery directly to the defect’s anatomical anchor points, providing a deterministic guide for inseting and suturing.

To fabricate these complex, patient-specific designs, the method had to satisfy stringent technical and safety requirements: high dimensional accuracy for precise registration, a smooth surface finish for atraumatic tissue contact, and sufficient rigidity to prevent deformation during handling. For eventual patient use, the material must also be certified as biocompatible and

capable of withstanding standard sterilization protocols. Based on these criteria, Stereolithography (SLA) 3D printing was selected for its sub-millimeter accuracy, smooth surface finish, and the availability of certified biocompatible photopolymer resins.

2.3. Benchtop Study

Patient-specific tongue phantoms were fabricated for benchtop surgical simulation. A 3D-printed mold (Figure 1A), generated from the patient’s pre-operative MRI data, was used to cast four identical, 1:1-scale tongue phantom replicas from Ecoflex™ 00-10 platinum-cure silicone (Figure 1B). This choice aligns with established standards for oral surgical simulation, where Ecoflex™ 00-10 and 00-30 are often cited for favorable haptics during dissection and retraction [25, 26, 27]. Although the apparent modulus of Ecoflex™ 00-10 (≈ 45 kPa [30]) exceeds values reported for a passive, fully relaxed tongue (1.6–12.2 kPa [31, 32, 33]), this choice addresses the discrepancy between synthetic matrices and highly non-linear, hyperelastic biological tissue. Native tongue tissue strain-hardens under the large deformations of surgical manipulation; to approximate this increased handling impedance without fibrous reinforcement, a modestly higher baseline stiffness is needed. Pilot testing also confirmed that softer alternatives (e.g., Ecoflex™ Gel) were unsuitable due to gravitational slumping and immediate suture pull-out. Accordingly, consistent with literature recommendations for surgical simulation, Ecoflex™ 00-10 was selected as an optimal trade-off: it provides structural integrity and suture retention without tearing, while remaining compliant enough to preserve the deformability challenges.

To provide a stable and anatomically relevant jig for the surgical simulation, a patient-specific jaw model was created (Figure 1C). The initial mandibular anatomy was derived from the patient’s MRI data; however, due to its low resolution, a high-resolution generic jaw model was elastically registered to the MRI-derived model using anatomical landmarks. The resulting jaw model was engineered as a rigid base to which the tongue phantom could be attached. A second key feature was its two-part construction, implemented to enable unobstructed 3D surface scanning for post-operative quantitative evaluation. The mandible was partitioned so that a stationary lower section provides all anatomical attachment points, while a removable upper section—anatomically independent of the tongue—was affixed to it. During 3D scanning, removal of the upper section provides an unobstructed view of the tongue’s entire superior surface, which is crucial for surface scanning.

A multi-layered phantom was fabricated to simulate the anatomy of the radial forearm free flap (RFFF) donor site (Figure 3D). The RFFF was selected for this benchtop demonstration because the simulated defects involve the oral tongue, a region where maximizing pliability and minimizing tethering are critical [34, 35, 36]. Although this study uses the RFFF, the toolkit’s architecture is adaptable to other donor sites, such as the anterolateral thigh (ALT). In the RFFF model, volume was controlled via the beavertail modification, in which additional subcutaneous adipose tissue (SAT) is harvested beyond the skin paddle and tucked to restore bulk. In contrast, applying this approach to an ALT flap would control volume through selection of the fascial elevation plane to set baseline thickness, followed by trimming adipose tissue to correct excess bulk [37, 23].

To simulate reconstruction with a beavertail RFFF, the phantom comprised a 1.5-mm-thick superficial skin layer laminated onto a 2.0-mm-thick SAT layer. The skin simulant was formulated following the methods of Chattrairat et al. [24], using an 88:12 weight ratio of Mold Star™ 30 and Ecoflex™ 00-10 platinum-cure silicones. This composite was selected to replicate the mechanical and haptic properties of human skin during surgical manipulation. The SAT layer was cast from Ecoflex™ 00-05 silicone, chosen for its low shear modulus to mimic the compliance and handling characteristics of human adipose tissue.

Beavertail SAT phantoms were also fabricated to provide the bulk adjustment required for benchtop reconstructions. Consistent with the clinical procedure in which subdermal fat and fascia are harvested, these phantoms were modeled as a single layer of adipose tissue simulant. To achieve the case-specific volume required by the virtual plan, each SAT phantom was cast from Ecoflex™ 00-05 silicone at a uniform thickness of 2.0 mm, with bulk adjusted accordingly (Figure 3F). Although a uniform thickness was used here, a clinical workflow could incorporate pre-operative MRI/CT of the patient’s forearm to quantify and map the available SAT for harvest. For this feasibility study, the planned flap volume matched the planned resection volume (i.e., 0% overbulking), as the phantom models do not simulate postoperative tissue atrophy.

Benchtop surgical simulation was conducted in a controlled laboratory setting. Each of the four tongue phantoms underwent the complete guided reconstruction using the patient-specific surgical toolkit. Figures 2 and 3 show the workflow for case C. To avoid bias from the senior surgeon who advised on the toolkit design, the procedure was performed by a different

co-author surgeon with no prior experience in this domain. This isolates toolkit performance by attributing reconstruction accuracy to the toolkit’s guidance rather than pre-existing surgical skill. The flap was secured to the tongue phantom with 4-0 silk braided sutures (Isaac Technologies, IC-SN4002, Canada). Suturing used a simple interrupted technique, placing stitches ≈ 5 mm apart and ≈ 3 mm from the wound edge. All suturing was performed by the same operator to minimize variability.

2.4. Quantitative Evaluation of Anatomical Accuracy

To quantitatively assess geometric fidelity, high-resolution 3D surface scans of each tongue phantom were acquired before the procedure and after the final reconstruction. For each case, the resulting pre-operative and post-operative 3D models were digitally aligned using an iterative closest point (ICP) algorithm with 100 iterations [38]. To ensure repeatable registration, ICP was applied exclusively to the static jaw model surface, which remained unchanged across scans; the resulting spatial transformation was then applied to the entire post-operative scan.

Maximum deviation, defined using the Hausdorff distance between the aligned pre-operative and post-operative models, was selected as the primary measure of anatomical accuracy, consistent with prior studies [39, 5, 40]. The Hausdorff distance captures the greatest discrepancy between two surfaces and therefore the worst-case geometric error. However, the absolute maximum is highly sensitive to outliers, which are common 3D scanning artifacts (e.g., specular reflections from the moist silicone surface, edge noise, and data voids in complex geometries). To enable a robust and clinically meaningful assessment of toolkit performance, the 99th percentile of surface deviation was used instead of the absolute maximum, mitigating the influence of a small number of extreme, non-representative errors. The PyVista library was used for these calculations and to generate surface deviation color maps for visualization.

3. Results

The proposed additively manufactured toolkit was used to perform a complete guided reconstruction on all four patient-specific tongue phantoms. During the surgical simulation, all components were used to guide their corresponding procedural steps, and the operator reported that the workflow

was straightforward and the guides were easy to handle. The final sutured reconstructions for all four cases are shown in Figure 4.

Post-operative 3D surface scans were compared with the pre-operative scans. The mean surface deviation across all four reconstructions ranged from 0.76 mm to 1.26 mm. The 99th percentile of surface deviation, a robust measure of maximum error, ranged from 3.37 mm to 4.13 mm. Detailed metrics for each case are presented in Table 1. The surface deviation color maps in Figure 5 provide a visual representation of these measurements. The maps show broad areas of low deviation (purple and blue), indicating close correspondence to the virtual plan across most of the reconstructed surface. Higher deviations (green and yellow) were consistently localized along the suture lines at the flap–tongue interface and at resection regions with high surface curvature.

Table 1: Summary of surface deviation metrics for the four reconstructed tongue phantoms

Case	Mean Deviation (mm)	99th Percentile Deviation (mm)
A	0.76	3.37
B	1.26	4.13
C	0.99	3.91
D	1.18	3.54

4. Discussion

These findings establish the feasibility of an additively manufactured surgical toolkit for guided tongue reconstruction and show that such a system can bridge the gap between VSP and its physical execution on a deformable soft-tissue phantom. A patient-specific toolkit was designed and successfully exercised in a complete end-to-end benchtop simulation. Qualitative assessment was particularly informative: although the operator had no prior involvement in the VSP, the workflow was reported as easy to understand, intuitive, and predictable. Each component fulfilled a distinct role: the resection guide conformed snugly for stable surface marking; the physical models provided spatial appreciation of target geometries; and the harvesting guides afforded definitive control over flap shape, volume, and accurate inseting for suturing. Complementing this subjective assessment, the quantitative evaluation confirmed high anatomical accuracy: achieved surface deviations were comparable to the high-fidelity results reported for guided os-

seous VSP, where mandibular reconstruction deviations frequently fall within 0.1–3.8 mm [5, 41]. Achieving this precision on a mobile soft-tissue model underscores the toolkit’s potential to translate a sophisticated virtual plan into a physically actionable reconstruction with high fidelity. Nonetheless, validation in clinical trials beyond the benchtop setting remains a necessary next step.

The proposed surgical toolkit is grounded in the proven VSP paradigm for mandibular and maxillofacial reconstruction; it adopts shared principles of high-fidelity plan translation, comprehensive guide systems, and intraoperative inspections [5, 7]. However, its design diverges to address the distinct objectives of tongue reconstruction. Whereas osseous reconstruction is a predominantly geometric task concerned with restoring a static structural framework and judged largely by spatial conformity [5, 12], tongue reconstruction is a biomechanical challenge aimed at restoring dynamic function (speech and swallowing) by providing sufficient bulk and mobility [42, 31, 43, 44]. This shift in goals necessitates reimagining guidance: bone guides exploit lock-and-key registration on rigid and landmark-rich surfaces [5, 45], but the tongue’s deformable anatomy precludes rigid fixation [13]. This distinction also extends to the nature of guidance. While osseous toolkits often provide active guidance by physically constraining an instrument’s trajectory, such forceful constraints are impractical and potentially traumatic for delicate soft tissue. Consequently, our toolkit shifts the paradigm to a system of referential and inspection guidance, designed to inform the surgeon’s judgment rather than rigidly control the instrument.

Moving beyond freehand execution—where the resection is guided solely by the surgeon’s visual interpretation of the VSP [20, 21]—our system provides three instruments to enhance accuracy and confidence. First, the Resection Marking Guide defines a physical boundary for the surface incision. Building on pioneering work with soft silicone guides [22], our design incorporates major tongue landmarks denoted on the guide itself to enable precise, repeatable registration. The guide can be secured with sutures to prevent slippage during marking. Second, the Resection Model, a 1:1 scale replica of the planned resection geometry, serves as a physical reference. It helps the surgeon visualize the complex resection geometry and estimate the required cut depth. Third, the Resection Volume Inspection Guide (Figure 2F) provides semi-quantitative feedback linking the resection and harvesting stages. Unlike purely visual estimation, this physical gauge flags deviations from the virtual plan: a resection specimen that prevents mold closure indicates a re-

section volume greater than planned, necessitating a proportionally greater flap volume to be harvested, and vice versa.

Although the current system effectively indicates the direction of deviation, it still relies on the surgeon’s judgment to estimate the magnitude of the required adjustment. To address the rigidity of pre-printed guides in the setting of unexpected tumor spread, we propose a hybrid workflow in future iterations. The toolkit would serve as a baseline map; if margins expand, rapid intraoperative scanning could digitize the de facto resection and algorithmically update the flap’s skin paddle design and volume requirements. This approach would convert the static toolkit into an adaptive system capable of informing necessary adjustments.

Guiding donor free flap dissection is the most critical phase of the reconstructive workflow, as its design and precise execution affect functional outcomes. The challenge is twofold: first, the flap must possess optimal planar geometry to reduce tongue tethering and improve mobility [31, 46, 47, 48, 49, 50]; second, it must provide adequate volume for proper palatal and oropharyngeal contact, which is essential for bolus propulsion, swallowing, and aspiration prevention [46, 20, 44].

Recognizing these imperatives, recent VSP approaches have evolved beyond landmark-based measurements [20, 21] to algorithmically design biomechanically optimized, patient-specific flaps that match the defect [10]. However, executing this virtual plan remains a significant challenge. One VSP technique uses 3D-printed flap-cutting guides to delineate the free flap’s planar shape and has demonstrated clinical benefits [8]. While this method offers excellent control over planar geometry, it provides no guidance on volume; the surgeon must still estimate the required SAT freehand. An alternative workflow addresses volumetric control by using a negative mould of the resected tissue as a guide; after harvesting, the flap is placed into this mould to assess and adjust its bulk until it matches the resected tissue [22]. By enforcing a 1:1 volume match to the resected tissue, however, this method cannot accommodate clinical strategies such as planned overbulking.

The proposed surgical toolkit synthesizes these approaches and adds an adaptive feedback mechanism to address key limitations. A key innovation is a decoupled harvesting system that provides independent control over both flap parameters. It uses a separate Planar Guide (Figure 3A) to outline the skin paddle, analogous to the CARST method [8], and a dedicated Volumetric Guide (Figure 3B) for precise bulk management. The Volumetric Guide also enables planned overbulking using a VSP-precalculated target volume.

This overbulking is necessary to counteract postoperative atrophy, which can range from 20% to over 50% and is often exacerbated by radiotherapy [51, 48, 52].

Although demonstrated here using the beavertail RFFF, this decoupled guiding architecture is adaptable to the ALT flap. In an ALT workflow, the volumetric guide would dictate the requisite fascial elevation plane. However, the practical implementation of intraoperative volume feedback differs. While the RFFF beavertail offers tunability via trimming of the beavertail tissue, adjusting ALT volume post-harvest typically requires delicate thinning of subcutaneous fat [23]. This process entails a higher risk of vascular compromise without precise perforator mapping. Future studies should assess this workflow for thicker, less pliable ALT flaps.

Despite its promising accuracy, the toolkit faces key limitations in translating a virtual plan to patients. A primary challenge is limited active guidance for resection depth and angle: the marking guide delineates the surface incision line but relies on the surgeon’s 3D spatial reference to a separate model to judge depth and angle. Additionally, the phantom cannot replicate the dynamic challenges of live surgery, such as bleeding, saliva, restricted access, and greater freedom of movement; consequently, achieving stable guide registration on a slippery organ will likely be more difficult than in this controlled setting. To improve registration in this environment, future guides should integrate suction channels for fluid evacuation and negative-pressure adhesion and incorporate temporary stay sutures for mechanical locking.

Methodological limitations, including the lack of a control group and reliance on a single operator, limit assessment of inter-surgeon variability, generalizability, and direct comparison with the standard of care. Future work requires controlled, multi-surgeon studies across a wider range of reconstruction cases [12, 8]. These studies should also incorporate qualitative face-validity assessments in which experienced surgeons perform the simulated surgery and rate the realism of tissue incision and suturing.

The fidelity of the silicone phantoms, while adequate for this feasibility study, remains a limitation. The homogeneous silicone simplifies the tongue’s complex layered, anisotropic, and viscoelastic properties [53, 54, 55]. Regardless of formulation, synthetic analogs lack the stress relaxation, biological heterogeneity, and high nonlinearity of real tissue. Future work should use high-fidelity, multi-material phantoms and quantitative sensitivity analyses across silicone grades to determine when specific material choices are necessary [56]. Additionally, the absence of biological atrophy precluded testing a

critical clinical capability of the volumetric guide: executing a pre-planned “overbulking” strategy to compensate for expected tissue loss.

Finally, while this study demonstrates high geometric accuracy, this endpoint is a surrogate for the true clinical goals of tongue reconstruction: improved speech, swallowing, and quality of life. The relationship between geometric fidelity and functional outcomes remains to be established. The ultimate goal is a prospective clinical trial comparing guided versus conventional reconstruction, but several practical hurdles must first be addressed, including validating that sterilization protocols do not compromise guide dimensional accuracy and evaluating the end-to-end turnaround time for VSP and manufacturing.

Conclusion

A patient-specific, additively manufactured surgical toolkit translated a sophisticated virtual tongue resection–reconstruction plan into a high-fidelity physical outcome in soft-tissue phantoms. It decouples flap’s planar geometry from its volumetric control and adds an intraoperative resection-volume inspection step. Together, these features create an adaptive workflow linking what is resected to how bulk is restored. This architecture transforms flap design from sequential estimation into a controllable process in which surface shape and tissue volume are specified independently, inspected, and reconciled. The workflow advances translation fidelity for tongue reconstruction VSP and provides a reproducible structure for a procedure that has historically relied on intraoperative estimation.

Current validation is limited by a homogeneous, non-biological phantom, a single operator, and the inability to model dynamic surgical conditions or postoperative atrophy. Even so, the results reflect plan execution accuracy on a tongue phantom. Next, a staged progression through multi-surgeon early clinical case studies will evaluate registration robustness, functional correlates of geometric accuracy, and the efficacy of planned overbulking, sterilization, and turnaround workflows.

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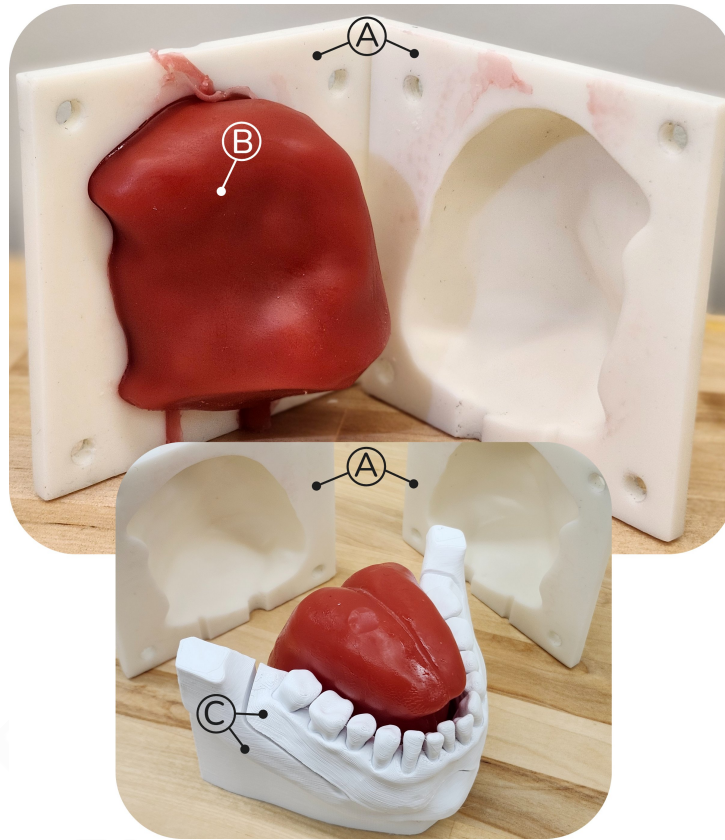


Figure 1: Fabrication of the soft silicone tongue phantom. (A) Two-part 3D-printed mold with alignment and fastening bosses. (B) Cast, pigmented elastomer tongue phantom after demolding. (C) Final positioning of the phantom within the 3D-printed jaw model.

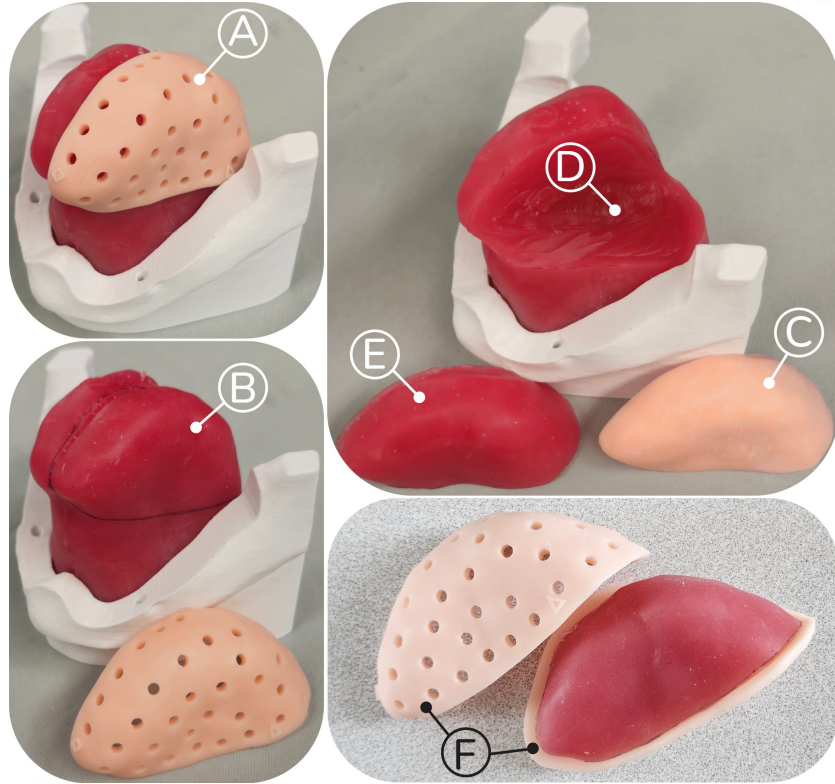


Figure 2: Patient-specific resection guidance toolkit and workflow. (A) Resection Marking Guide positioned on the tongue phantom to delineate the surface incision. (B) Marked resection boundary on the tongue surface using the conforming guide. (C) Solid Resection Model providing a physical reference for 3D cut depth and contour. (D) Completed resection. (E) Excised tissue specimen ready for volumetric inspection. (F) Resection Volume Inspection Guide with the excised tissue positioned for intraoperative volume verification against the planned resection.

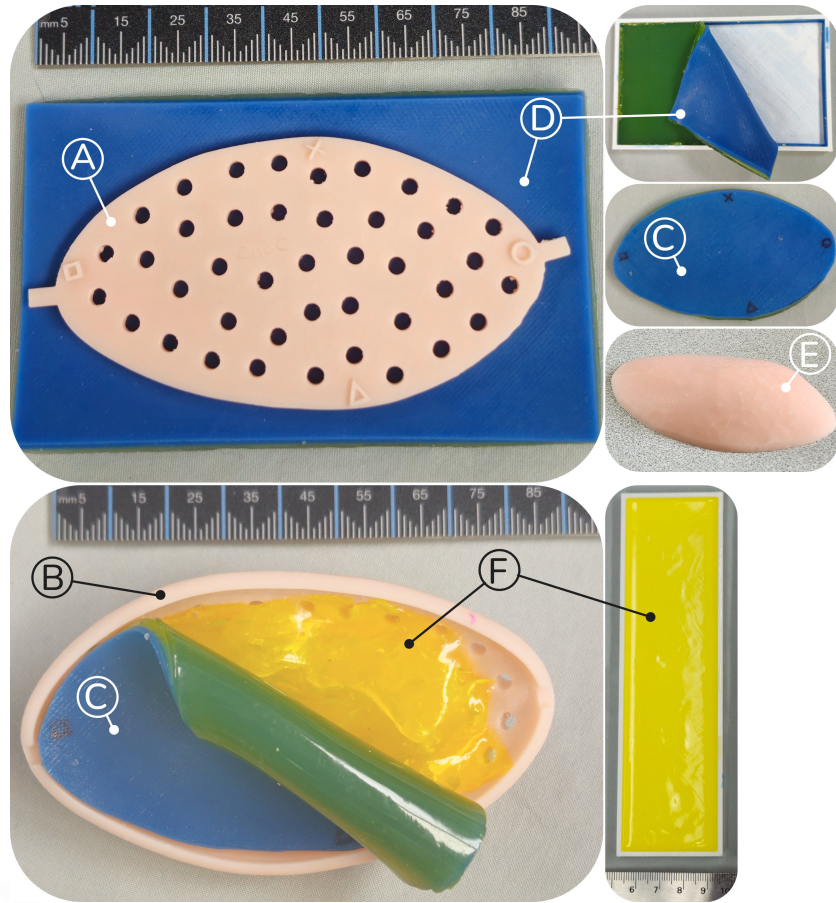


Figure 3: Patient-specific flap-harvesting guidance toolkit and workflow. (A) Planar template guide to outline skin-perimeter geometry on the donor-site phantom. (B) Volumetric guide for verifying beavertail SAT volume against planned requirements. (C and D) Multi-layer donor-site phantom with skin and SAT layers. (E) 3D-printed Flap Model providing a physical reference for target volume and shape. (F) Harvested beavertail SAT positioned in the volumetric guide for intraoperative volume verification.

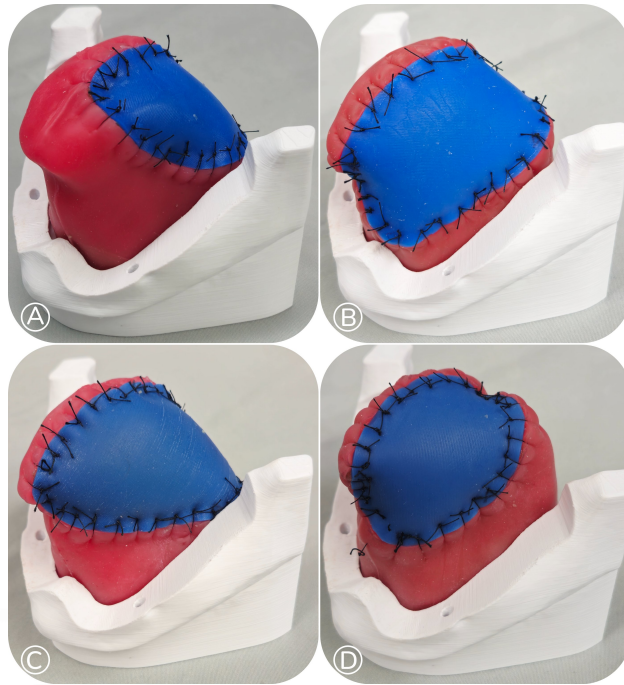


Figure 4: Final reconstructed tongue phantoms for the four VSP cases. Each phantom demonstrates successful translation of the virtual surgical plan to physical execution using the patient-specific toolkit.

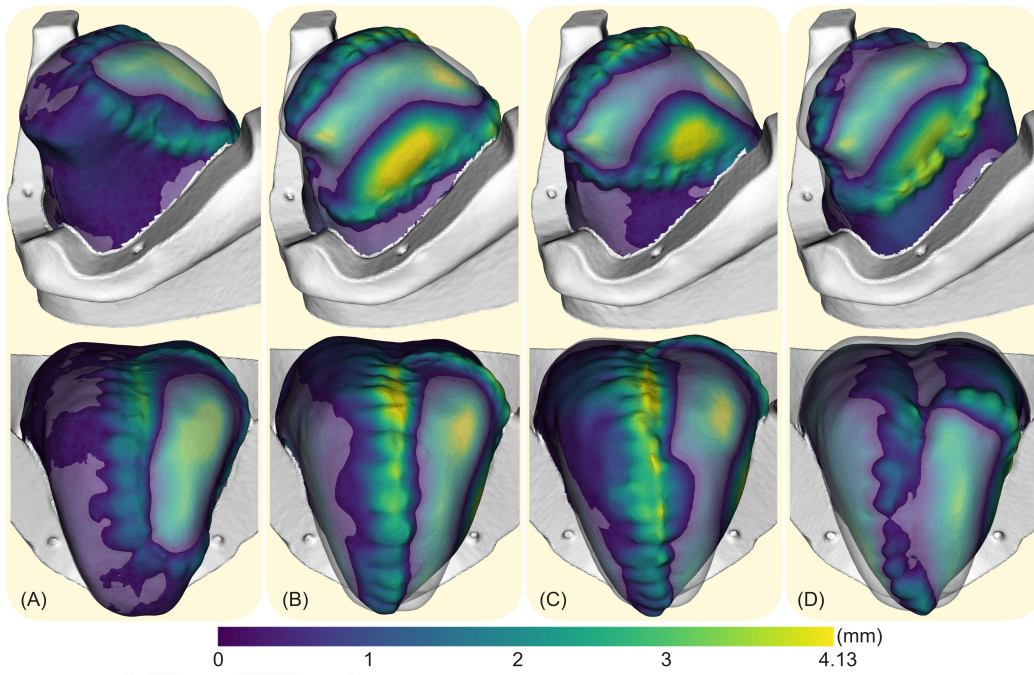


Figure 5: Post-reconstruction surface accuracy for the four simulated cases (A–D). Color map (0–4.13 mm) shows pointwise surface deviation between the executed reconstruction and the reference pre-operative tongue model.